

Dimensions polishing up

bachotex 2025 meeting

What are dimensions

```
\scratchdimen = 10pt
```

10.0pt

```
\scratchdimen 10pt
```

10.0pt

```
\scratchdimen \lineheight
```

17.3281pt

```
\scratchdimen 2em
```

23.98682pt

```
\scratchdimen - \numexpr 65536 * 4 + 65536 / 4 \relax sp
```

-4.25pt

properties: direct, units, accuracy, indirect, font related, just numbers

Simple ε -T_EX expressions

```
\scratchdimen = \dimexpr 10pt + 1ex \relax
```

16.1886pt

```
\def\whatever#1{\scratchdimen = \dimexpr 10pt + #1 \relax}
```

```
\whatever{1ex}
```

```
\whatever{\dimexpr 1ex}
```

16.1886pt

```
\scratchdimen = 33.3pt \divide\scratchdimen 33
```

1.00908pt

```
\scratchdimen = \dimexpr 33.3pt / 33\relax
```

1.0091pt

properties: order matters, read on as long as valid, just +-/*, incompatible division

Simple ε -T_EX expressions in LuaMetaT_EX

```
\scratchdimen = 33.3pt \divide\scratchdimen 33
```

1.00908pt

```
\scratchdimen = \dimexpr 33.3pt : 33\relax
```

1.00908pt

properties: also div and mod : ; (only enabled in ConT_EXt)

Extended ε -T_EX expressions

```
\scratchdimen = \dimexpression {10pt + 1ex}
```

16.1886pt

properties: bounded by braces, more operations, order matters less, compromise, dimensions, floats and integers

!	not	negate	<	less
\mp	nmp	negative	>	greater
\pm	npm	positive	<> \neq	not equal
+		add	<= \leq	less equal
-		subtract	>= \geq	greater equal
*		multiply	&& \and	conditional and
/		divide	\or	conditional or
:	div	idivide	and	and
;	% mod	imodulo	or	or
=		equal		

Expression assignments

```
\scratchdimen = {10pt + 1ex}
```

16.1886pt

```
\def\whatever#1{%
```

```
  \scratchdimen = \dimexpr 10pt + {#1} \relax
```

```
}
```

```
\whatever{1ex / 4.25}
```

11.45615pt

```
\scratchdimen{10pt + 1ex}
```

16.1886pt

```
\scratchdimen\dimexpr {10pt + 1ex} \relax
```

16.1886pt

More tolerant units

```
\scratchdimen{10pt + ex}
```

16.1886pt

```
\uunit 10pt \scratchdimen{10pt + 2uu}
```

30.0pt

```
\newdimension\xxunit \newuserunit \xxunit xx
```

```
\xxunit 20pt \scratchdimen{10pt - xx}
```

-10.0pt

```
\scratchcounter\dimexpression{(lh == (sh + sd))}
```

1

Order matters less

```
\scratchdimen{10pt * 2}
```

20.0pt

```
\scratchdimen{2 * 10pt}
```

20.0pt

Better integration

```
\scratchdimen{2.1 * 10pt + ex/2 + 1h}
```

41.42245pt

```
\framed[height=2lh+ex,width=tw/2]{indeed}
```

indeed

properties: efficient (neglectable overhead), no explicit expression clutter

Going further

```
\scratchdimen\dimexperimental{2.1 * 10pt + ex/2 + 1h}
```

41.42245pt

Going Polish

```
[dimexpression rpn 1 s: 2.1 10.0pt * 6.1886pt 2 / + 17.3281pt +]
[dimexpression rpn 1 r: ]
[dimexpression rpn 2 s: 10.0pt * 6.1886pt 2 / + 17.3281pt +]
[dimexpression rpn 2 r: 2.1]
[dimexpression rpn 3 s: * 6.1886pt 2 / + 17.3281pt +]
[dimexpression rpn 3 r: 2.1 10.0pt]
[dimexpression rpn 4 s: * 6.1886pt 2 / + 17.3281pt +]
[dimexpression rpn 4 r: 21.00006pt]
[dimexpression rpn 5 s: 2 / + 17.3281pt +]
[dimexpression rpn 5 r: 21.00006pt 6.1886pt]
[dimexpression rpn 6 s: / + 17.3281pt +]
[dimexpression rpn 6 r: 21.00006pt 6.1886pt 2]
[dimexpression rpn 7 s: / + 17.3281pt +]
[dimexpression rpn 7 r: 21.00006pt 3.0943pt]
[dimexpression rpn 8 s: / + 17.3281pt +]
[dimexpression rpn 8 r: 24.09436pt]
[dimexpression rpn 9 s: +]
[dimexpression rpn 9 r: 24.09436pt 17.3281pt]
```

Not only values

```
\ifboolean \dimexperimental{(ex > em) and (lh > 2ex)} Yes \else No \fi  
No
```

Normalizing types

```
[dimexpression rp1 1 s: 6.1886pt 11.99341pt > 17.3281pt 12.3772pt > and]
[dimexpression rp1 1 r: ]
[dimexpression rp2 2 s: 11.99341pt > 17.3281pt 12.3772pt > and]
[dimexpression rp2 2 r: 6.1886pt]
[dimexpression rp3 3 s: > 17.3281pt 12.3772pt > and]
[dimexpression rp3 3 r: 6.1886pt 11.99341pt]
[dimexpression rp4 4 s: > 17.3281pt 12.3772pt > and]
[dimexpression rp4 4 r: 0]
[dimexpression rp5 5 s: 12.3772pt > and]
[dimexpression rp5 5 r: 0 17.3281pt]
[dimexpression rp6 6 s: > and]
[dimexpression rp6 6 r: 0 17.3281pt 12.3772pt]
[dimexpression rp7 7 s: > and]
[dimexpression rp7 7 r: 0 1]
```

Cheating

```
\thewithoutunit    \dimexperimental{2.1 * 4.2}  
\todimension       \dimexperimental{2.1 * 4.2}  
\toscaled          \dimexperimental{2.1 * 4.2}  
\tosparsedimension\dimexperimental{2.1 * 4.2}  
\tosparsescaled    \dimexperimental{2.1 * 4.2}  
\the               \floatexpr2.1 * 4.2\relax  
\tolimitedfloat    \floatexpr2.1 * 4.2\relax
```

8.82002

8.82002pt

8.82002

8.82002pt

8.82002

8.8199999332427978516

8.82